WORKED SOLUTION

STEP III 2020, Question 1

Pure • Reduction Formulae

STEP III 2020, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Part (i) is a reduction formula, so integration by parts is the natural first move: differentiate the power of $\cos x$ and integrate $\cos bx$. That leaves a product $\sin x \sin bx$, and the compound-angle formula for $\cos(b-1)x$ turns it back into integrals of the original type. Part (ii) then needs nothing new: the recurrence lowers both a and b by one, which keeps the gap $b-a = 2m+1$ fixed, so it steps neatly from n to $n+1$.

Solution

Part (i)

Let $a, b \geq 1$. Integrate by parts with $u = \cos^a x$ and $\mathrm{d}v = \cos bx\,\mathrm{d}x$, so that $\mathrm{d}u = -a\cos^{a-1}x \sin x\,\mathrm{d}x$ and $v = \frac{1}{b}\sin bx$ (fine, since $b \neq 0$):

$$\mathrm{I}(a,b) = \left[\frac{\cos^a x \sin bx}{b}\right]_0^{\pi/2} + \frac{a}{b}\int_0^{\pi/2} \cos^{a-1} x \sin x \sin bx\,\mathrm{d}x.$$

The boundary term vanishes at both ends: at $x = \frac{\pi}{2}$ because $\cos^a \frac{\pi}{2} = 0$ (this is where $a \geq 1$ is used), and at $x = 0$ because $\sin 0 = 0$.

Ask yourself How can we turn $\sin x \sin bx$ back into cosines of multiples of x?

We want $\mathrm{I}(a-1, b-1)$ to appear, so look at $\cos(b-1)x$:

$$\cos(b-1)x = \cos bx \cos x + \sin bx \sin x \quad \Longrightarrow \quad \sin x \sin bx = \cos(b-1)x - \cos x \cos bx.$$

Substituting,

$$\begin{aligned}\mathrm{I}(a,b) &= \frac{a}{b}\int_0^{\pi/2} \cos^{a-1} x \cos(b-1)x\,\mathrm{d}x - \frac{a}{b}\int_0^{\pi/2} \cos^a x \cos bx\,\mathrm{d}x \\ &= \frac{a}{b}\mathrm{I}(a-1,b-1) - \frac{a}{b}\mathrm{I}(a,b).\end{aligned}$$

Multiplying by b and collecting the $\mathrm{I}(a,b)$ terms gives $(a+b)\,\mathrm{I}(a,b) = a\,\mathrm{I}(a-1,b-1)$, and since $a+b>0$,

$$\mathrm{I}(a,b) = \frac{a}{a+b}\,\mathrm{I}(a-1,b-1),$$

as required.

Part (ii)

The integral on the left is $\mathrm{I}(n,\, n+2m+1)$. Write

$$R(n,m) = (-1)^m\,\frac{2^n\,n!\,(2m)!\,(n+m)!}{m!\,(2n+2m+1)!}$$

for the right-hand side. Fix any non-negative integer m; we prove $\mathrm{I}(n, n+2m+1) = R(n,m)$ for all $n \geq 0$ by induction on n.

Base case, $n=0$.

$$\mathrm{I}(0,2m+1) = \int_0^{\pi/2} \cos(2m+1)x\,\mathrm{d}x = \frac{\sin((2m+1)\frac{\pi}{2})}{2m+1} = \frac{(-1)^m}{2m+1},$$

because $\sin(m\pi+\frac{\pi}{2}) = \cos m\pi = (-1)^m$. On the other side,

$$R(0,m) = (-1)^m\,\frac{1\cdot 1\cdot(2m)!\,m!}{m!\,(2m+1)!} = \frac{(-1)^m}{2m+1}.$$

So the result holds for $n=0$.

Inductive step. Suppose $\mathrm{I}(k, k+2m+1) = R(k,m)$ for some $k \geq 0$. Apply part (i) with $a = k+1$ and $b = k+2m+2$, both positive integers, so that $a-1=k$, $b-1 = k+2m+1$ and $a+b = 2k+2m+3$:

$$\mathrm{I}(k+1,\, k+2m+2) = \frac{k+1}{2k+2m+3}\,\mathrm{I}(k,\, k+2m+1) = \frac{k+1}{2k+2m+3}\,R(k,m).$$

Note that $k+2m+2 = (k+1)+2m+1$, so the left-hand side is exactly the integral for $n = k+1$ with the same m.

> **Ask yourself** What does the formula say $R(k+1,m)/R(k,m)$ should be?

Going from $n=k$ to $n=k+1$ in R, the factor $(-1)^m(2m)!/m!$ is unchanged, and

$$\begin{aligned}\frac{R(k+1,m)}{R(k,m)} &= \frac{2^{k+1}}{2^k}\cdot\frac{(k+1)!}{k!}\cdot\frac{(k+m+1)!}{(k+m)!}\cdot\frac{(2k+2m+1)!}{(2k+2m+3)!}\\ &= \frac{2(k+1)(k+m+1)}{(2k+2m+3)(2k+2m+2)} = \frac{k+1}{2k+2m+3},\end{aligned}$$

using $2k+2m+2 = 2(k+m+1)$. Hence

$$\mathrm{I}(k+1,\, k+2m+2) = \frac{k+1}{2k+2m+3}\,R(k,m) = R(k+1,m),$$

which is the statement for $n = k + 1$.

Since the result holds for $n = 0$, and its truth for $n = k$ implies its truth for $n = k + 1$, it holds for all non-negative integers n, by induction. As m was arbitrary, this holds for all non-negative integers m too:

$$\int_0^{\pi/2} \cos^n x \, \cos(n + 2m + 1)x \, \mathrm{d}x = (-1)^m \frac{2^n \, n! \, (2m)! \, (n + m)!}{m! \, (2n + 2m + 1)!}.$$

Checks

- Take $n = 1$, $m = 0$: the formula gives $\frac{2 \cdot 1 \cdot 1 \cdot 1}{1 \cdot 3!} = \frac{1}{3}$. Directly, $\cos x \cos 2x = \frac{1}{2}(\cos 3x + \cos x)$ integrates over $[0, \frac{\pi}{2}]$ to $\frac{1}{2}\left(-\frac{1}{3} + 1\right) = \frac{1}{3}$.
- Take $n = 1$, $m = 1$: the formula gives $-\frac{2 \cdot 2 \cdot 2}{5!} = -\frac{1}{15}$, and a numerical evaluation of $\int_0^{\pi/2} \cos x \cos 5x \, \mathrm{d}x$ gives -0.066667.
- The recurrence in (i) and the closed form in (ii) were also confirmed by computer for all $n \leq 6$, $m \leq 5$.

Common slips

- Forgetting to say why the boundary term vanishes; the top limit needs $a \geq 1$.
- Inducting on n but letting m change: part (i) lowers a and b together, so m stays fixed.
- Sign errors in $\sin\left((2m + 1)\frac{\pi}{2}\right)$; it alternates $1, -1, 1, \ldots$, giving the $(-1)^m$.

Beyond STEP

Reduction formulae like this are the integral analogue of recurrence relations, and the closed form is built from ratios of factorials in the same way as Wallis's product and the Beta function $\int_0^{\pi/2} \cos^p x \sin^q x \, \mathrm{d}x$, which you meet early in a university analysis or methods course. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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